

Toroidal vesicle geometry and equilibrium

These notes reconstruct the calculations in *Physical Review A* **41**, 4517–4520 (1990) [1]. Numbers (1)–(23) identify the original equations; lettered tags give intermediate calculations or explicit checks. The symbols and curvature convention follow the paper. The second-variation formula used below is precisely the 1989 formula cited by that paper [2]. No later vesicle models or additional physical examples are included. The exact equilibrium calculation is reproduced; the printed stability argument has discrepancies, which are identified where they occur rather than assumed away.

1 Toroidal geometry and the exact equilibrium

The constrained shape energy and its equilibrium equation are

$$F = \frac{k_c}{2} \oint (c_1 + c_2 - c_0)^2 dA + \Delta p \int dV + \lambda \oint dA, \quad \Delta p = p_o - p_i. \quad (1)$$

$$\Delta p - 2\lambda H + 4k_c(H + \tfrac{1}{2}c_0)(H^2 - K - \tfrac{1}{2}c_0 H) + 2k_c \nabla^2 H = 0. \quad (2)$$

Here k_c is the bending modulus, c_0 the spontaneous curvature, λ the area multiplier, $H = -(c_1 + c_2)/2$, and $K = c_1 c_2$. The Laplacian is intrinsic to the surface. To see the energy-to-equation step explicitly, use the first-variation convention of (2):

$$\delta dA = -2H\psi dA, \quad \delta V = \oint \psi dA, \quad \delta H = \tfrac{1}{2}\nabla^2 \psi + (2H^2 - K)\psi. \quad (2a)$$

Since the curvature energy is $k_c(2H + c_0)^2/2$, its variation is

$$\delta F_c = k_c \oint [(2H + c_0)\nabla^2 \psi + (2H + c_0)(2H^2 - 2K - c_0 H)\psi] dA. \quad (2b)$$

Two integrations by parts on the closed surface give

$$\oint (2H + c_0)\nabla^2 \psi dA = \oint \psi \nabla^2 (2H + c_0) dA = 2 \oint \psi \nabla^2 H dA. \quad (2c)$$

Adding $\Delta p \delta V + \lambda \delta A$ and setting the coefficient of arbitrary ψ to zero produces (2). The volume and normal signs in this calculation are tied to (2a); changing the orientation requires changing them consistently. In particular, the toroidal parametrization below uses the paper's positive H at its outer equator.

The paper announces the stability bound

$$c_0 r_0 < -(\pi\sqrt{2})^{1/2}(\tfrac{3}{2} + \tfrac{\sqrt{2}}{4}) \simeq -3.9069. \quad (3)$$

Its derivation and limitations are examined in the next section. The geometric starting point is

$$\mathbf{Y} = \left(\frac{c \sinh \eta \cos \phi}{\cosh \eta - \cos \theta}, \frac{c \sinh \eta \sin \phi}{\cosh \eta - \cos \theta}, \frac{c \sin \theta}{\cosh \eta - \cos \theta} \right), \quad (4)$$

with $-\pi \leq \theta \leq \pi$, $0 \leq \phi < 2\pi$, and initially $c > 0$, $\eta > 0$. The positive length c fixes scale and η fixes the radius ratio. Direct elimination of θ gives

$$\left(\sqrt{x^2 + y^2} - c \coth \eta \right)^2 + z^2 = \frac{c^2}{\sinh^2 \eta}. \quad (4a)$$

For example, the numerator on the left, after subtracting the center, is

$$(\cosh \eta \cos \theta - 1)^2 + \sinh^2 \eta \sin^2 \theta = (\cosh \eta - \cos \theta)^2. \quad (4b)$$

Thus the generating-circle radius is $c/\sinh \eta$, its center is $c \coth \eta$ from the axis, and their ratio is $1/\cosh \eta$.

Here are the differentiations behind the curvature formulas. The tangent vectors are

$$\mathbf{Y}_\theta = \frac{c}{(\cosh \eta - \cos \theta)^2} (-\sinh \eta \sin \theta \cos \phi, -\sinh \eta \sin \theta \sin \phi, \cosh \eta \cos \theta - 1), \quad (4c)$$

$$\mathbf{Y}_\phi = \frac{c \sinh \eta}{\cosh \eta - \cos \theta} (-\sin \phi, \cos \phi, 0). \quad (4d)$$

Taking their scalar products and cross product yields

$$g_{\theta\theta} = \frac{c^2}{(\cosh \eta - \cos \theta)^2}, \quad g_{\phi\phi} = \frac{c^2 \sinh^2 \eta}{(\cosh \eta - \cos \theta)^2}, \quad g_{\theta\phi} = 0, \quad (4e)$$

The area density is $\sqrt{g} = c^2 \sinh \eta / (\cosh \eta - \cos \theta)^2$, and the unit normal is

$$\mathbf{n} = \frac{((1 - \cosh \eta \cos \theta) \cos \phi, (1 - \cosh \eta \cos \theta) \sin \phi, -\sinh \eta \sin \theta)}{\cosh \eta - \cos \theta}. \quad (4f)$$

With $L_{ij} = \mathbf{n} \cdot \mathbf{Y}_{ij}$ one obtains

$$L_{\theta\theta} = \frac{c \sinh \eta}{(\cosh \eta - \cos \theta)^2}, \quad L_{\phi\phi} = \frac{c \sinh \eta (\cosh \eta \cos \theta - 1)}{(\cosh \eta - \cos \theta)^2}, \quad L_{\theta\phi} = 0. \quad (4g)$$

The eigenvalues of $g^{-1}L$ are therefore $\sinh \eta / c$ and $(\cosh \eta \cos \theta - 1) / (c \sinh \eta)$. Their half-sum and product give H and K . Furthermore,

$$\nabla^2 = \frac{1}{\sqrt{g}} \partial_i (\sqrt{g} g^{ij} \partial_j) = \frac{(\cosh \eta - \cos \theta)^2}{c^2} \left(\partial_\theta^2 + \frac{\partial_\phi^2}{\sinh^2 \eta} \right). \quad (4h)$$

Both coefficients inside the divergence before multiplication by $1/\sqrt{g}$ are constant. Since H is independent of ϕ ,

$$H_\theta = -\frac{\coth \eta}{2c} \sin \theta, \quad H_{\theta\theta} = -\frac{\coth \eta}{2c} \cos \theta. \quad (4i)$$

Setting $c = 1$ gives the three curvature relations:

$$H = \frac{\sinh^2 \eta + K}{2 \sinh \eta}, \quad K = \cosh \eta \cos \theta - 1, \quad \nabla^2 H = -\frac{1}{2} \cos \theta (\cosh \eta - \cos \theta)^2 \coth \eta. \quad (5)$$

Lengths and c_0 are henceforth expressed in these units until scale is restored. The useful cancellation in the nonlinear term is

$$H^2 - K = \frac{\cosh^2 \eta (\cosh \eta - \cos \theta)^2}{4 \sinh^2 \eta}. \quad (5a)$$

Expanding (2) first in powers of c_0 gives

$$\Delta p - 2k_c c_0 K - (k_c c_0^2 + 2\lambda)H + 4k_c H(H^2 - K) + 2k_c \nabla^2 H = 0. \quad (5b)$$

Inserting (5) and (5a) into (5b) produces the original equation

$$\begin{aligned} 0 = & \Delta p - 2k_c c_0 (\cosh \eta \cos \theta - 1) - \frac{k_c c_0^2 + 2\lambda}{2} \left(\sinh \eta + \frac{\cosh \eta \cos \theta - 1}{\sinh \eta} \right) \\ & + \frac{k_c}{2} (\cosh \eta - \cos \theta)^2 \left[\frac{\cosh^2 \eta (\sinh^2 \eta - 1)}{\sinh^3 \eta} + \left(\frac{\cosh^3 \eta}{\sinh^3 \eta} - \frac{2 \cosh \eta}{\sinh \eta} \right) \cos \theta \right]. \end{aligned} \quad (6)$$

This is a polynomial identity in $\cos \theta$. Its cubic coefficient is

$$\frac{k_c \cosh \eta}{2 \sinh^3 \eta} (1 - \sinh^2 \eta) = 0. \quad (6a)$$

For a nondegenerate torus and $k_c > 0$, this requires

$$\sinh \eta = 1, \quad \cosh \eta = \sqrt{2}. \quad (7)$$

The entire last line of (6) then vanishes. Its remaining constant and linear terms are

$$\Delta p + 2k_c c_0 - \sqrt{2} (2k_c c_0 + \frac{1}{2} k_c c_0^2 + \lambda) \cos \theta = 0. \quad (7a)$$

Both coefficients vanish precisely when

$$\Delta p = -2k_c c_0, \quad \lambda = -2k_c (c_0 + \frac{1}{4} c_0^2). \quad (8)$$

Together (4a) and (7) give $(y - \sqrt{2})^2 + z^2 = 1$ in the half-plane $\phi = \pi/2$. This proves the radius ratio $1/\sqrt{2}$ within the family of circular standard tori considered by the paper. It is a stationarity result; stability still requires a second-variation analysis.

2 Second variation and scale

The deformed surface and Fourier expansion in the paper are

$$\mathbf{Y}' = \mathbf{Y} + \mathbf{n}\psi, \quad (9)$$

$$\psi = \sum_m [a_m(\theta) \sin m\phi + b_m(\theta) \cos m\phi]. \quad (10)$$

The amplitudes are smooth and periodic in θ . At (7), with $c = 1$, the geometry reduces to

$$g_{\theta\theta} = g_{\phi\phi} = (\sqrt{2} - \cos \theta)^{-2}, \quad H = \frac{\cos \theta}{\sqrt{2}}, \quad K = \sqrt{2} \cos \theta - 1. \quad (10a)$$

Consequently $\nabla^2 = (\sqrt{2} - \cos \theta)^2 (\partial_\theta^2 + \partial_\phi^2)$. The nonzero connection coefficients needed in the cited 1989 formula are

$$\Gamma_{\theta\theta}^\theta = \Gamma_{\theta\phi}^\phi = \Gamma_{\phi\theta}^\phi = -\frac{\sin \theta}{\sqrt{2} - \cos \theta}, \quad \Gamma_{\phi\phi}^\theta = \frac{\sin \theta}{\sqrt{2} - \cos \theta}. \quad (10b)$$

Also $L_{\theta\theta} = g_{\theta\theta}$ and $L_{\phi\phi} = K g_{\phi\phi}$. The following is (11) *as printed*, including its highest-derivative coefficient and omitted k_c :

$$\begin{aligned} \delta F = \pi \int_{-\pi}^{\pi} d\theta \sum_m \Big\{ & (a_{m,\theta\theta}^2 + b_{m,\theta\theta}^2) (\sqrt{2} - \cos \theta)^2 \\ & + (a_{m,\theta}^2 + b_{m,\theta}^2) [m^2 (\sqrt{2} - \cos \theta)^2 + \sqrt{2} (1 + c_0) \cos \theta - \frac{1}{2} \cos^2 \theta - 2 - 2c_0] \\ & + (a_m^2 + b_m^2) \left[m^2 \left[\frac{1}{2} m^2 (\sqrt{2} - \cos \theta)^2 + 3\sqrt{2} \cos \theta - \frac{1}{2} \cos^2 \theta \right] + (\sqrt{2} - \cos \theta)^{-2} [4 \cos^4 \theta \right. \\ & \left. + \sqrt{2} (c_0 - 9) \cos^3 \theta + (12 - 4c_0) \cos^2 \theta - \sqrt{2} (2 + c_0) \cos \theta + 2c_0] \right] \Big\}. \end{aligned} \quad (11)$$

Direct reduction of the formula cited for (11). For a reproducible comparison, δF below denotes the quadratic energy increment, with its Taylor factor 1/2 included, as in Eq. (39)

of the 1989 paper. Substitution of (8), (10a), and (10b) in that expression, before the final integrations by parts, gives

$$\begin{aligned} \frac{\delta F}{k_c} = & \int_{-\pi}^{\pi} d\theta \int_0^{2\pi} d\phi \left\{ \left[2 - (2\sqrt{2} + \frac{c_0}{\sqrt{2}}) \cos \theta + \frac{\sqrt{2}c_0}{\sqrt{2} - \cos \theta} \right] \psi^2 \right. \\ & - \left(2c_0 + \frac{1}{2} \cos^2 \theta + \sqrt{2} \cos \theta \right) \psi_\theta^2 - \left(\sqrt{2}c_0 \cos \theta + \frac{5}{2} \cos^2 \theta - \sqrt{2} \cos \theta \right) \psi_\phi^2 \\ & \left. - 2 \left[(\sqrt{2} + \frac{c_0}{\sqrt{2}}) \cos \theta - 1 \right] \psi(\psi_{\theta\theta} + \psi_{\phi\phi}) + \frac{1}{2} (\sqrt{2} - \cos \theta)^2 (\psi_{\theta\theta} + \psi_{\phi\phi})^2 \right\}. \end{aligned} \quad (11a)$$

Some component details make this substitution checkable. In the 1989 notation, L^{ij} is the matrix inverse of L_{ij} , so

$$\sqrt{g}(2Hg^{\theta\theta} - KL^{\theta\theta}) = 1, \quad \sqrt{g}(2Hg^{\phi\phi} - KL^{\phi\phi}) = K. \quad (11b)$$

The two differentiated contributions to the coefficient of ψ^2 , after including the area element, become $2H_{\theta\theta}$ and $-\sqrt{2}\partial_\theta[(H + c_0/2)\sin\theta]$. The gradient coefficients are obtained from

$$\frac{\lambda}{2k_c} + (H + \frac{1}{2}c_0)^2 + 2(H + \frac{1}{2}c_0)(K - 3H) = -2c_0 - \frac{1}{2} \cos^2 \theta - \sqrt{2} \cos \theta, \quad (11c)$$

and the same expression with K replaced by 1 gives the coefficient of ψ_ϕ^2 in (11a). The mixed term is $-2(K + c_0H)\psi(\psi_{\theta\theta} + \psi_{\phi\phi})$. Finally, $\frac{1}{2} \oint (\nabla^2 \psi)^2 dA$ gives exactly the last square.

For $m \geq 1$, either sine or cosine has squared integral π over ϕ ; the two sectors are orthogonal. For the sine sector, for example, $\psi_{\phi\phi} = -m^2\psi$. The necessary integrations over θ are

$$\int_{-\pi}^{\pi} a_m a_{m,\theta\theta} d\theta = - \int_{-\pi}^{\pi} a_{m,\theta}^2 d\theta, \quad (11d)$$

$$\int_{-\pi}^{\pi} \cos \theta a_m a_{m,\theta\theta} d\theta = - \int_{-\pi}^{\pi} \cos \theta a_{m,\theta}^2 d\theta - \frac{1}{2} \int_{-\pi}^{\pi} \cos \theta a_m^2 d\theta, \quad (11e)$$

$$\int_{-\pi}^{\pi} (\sqrt{2} - \cos \theta)^2 a_m a_{m,\theta\theta} d\theta = \int_{-\pi}^{\pi} \left[-(\sqrt{2} - \cos \theta)^2 a_{m,\theta}^2 + (\sqrt{2} \cos \theta - \cos 2\theta) a_m^2 \right] d\theta. \quad (11f)$$

There are no boundary terms because of periodicity. Expanding the square in (11a) and using (11d)–(11f) now yields

$$\begin{aligned} \frac{\delta F}{\pi k_c} = & \int_{-\pi}^{\pi} d\theta \left\{ \frac{1}{2} (\sqrt{2} - \cos \theta)^2 a_{m,\theta\theta}^2 \right. \\ & + \left[m^2 (\sqrt{2} - \cos \theta)^2 + \sqrt{2} (1 + c_0) \cos \theta - \frac{1}{2} \cos^2 \theta - 2 - 2c_0 \right] a_{m,\theta}^2 \\ & \left. + \left[m^2 \left(\frac{1}{2} m^2 (\sqrt{2} - \cos \theta)^2 - \frac{1}{2} \cos^2 \theta + 2\sqrt{2} \cos \theta - 3 \right) + \sqrt{2} \left(\sqrt{2} - \cos \theta + \frac{c_0}{\sqrt{2} - \cos \theta} \right) \right] a_m^2 \right\}. \end{aligned} \quad (11g)$$

The b_m sector has the same expression. For $m = 0$, only b_0 remains, and the prefactor is $2\pi k_c$, because the squared constant integrates to 2π . Thus the missing 1/2 in the first term of printed (11) cannot be repaired by a common normalization: the first-derivative coefficient already agrees, while the zero-derivative coefficient differs as well. Equation (11g) is the explicit reduction of the formula the paper cites, not an assertion that (11) has been independently reproduced.

What the printed stability argument establishes. The paper gives

$$c_0 < -(\frac{3}{2} + \frac{\sqrt{2}}{4}). \quad (12)$$

This bound does follow from positivity of the first-derivative coefficient in (11). Its m^2 term is nonnegative, and the remainder is concave as a function of $\cos \theta$. Consequently its minimum is at $\cos \theta = \pm 1$. The endpoint inequalities are

$$\sqrt{2} - \frac{5}{2} + (\sqrt{2} - 2)c_0 > 0, \quad -\sqrt{2} - \frac{5}{2} - (\sqrt{2} + 2)c_0 > 0. \quad (12a)$$

They respectively give $c_0 < -3/2 - \sqrt{2}/4$ and $c_0 < -3/2 + \sqrt{2}/4$, so the first is the stronger condition. However, the printed claim that (12) also makes the coefficient of $a_m^2 + b_m^2$ positive fails directly: at $m = 0$ and $\cos \theta = 0$ that coefficient in (11) is $c_0 < 0$.

There is also a simple check within the original perturbation family. Take $m = 1$ and a constant a_1 in (11g). Its zero-derivative bracket collapses to $\sqrt{2}c_0/(\sqrt{2} - \cos \theta)$, and hence

$$\delta F = 2\sqrt{2}\pi^2 k_c c_0 a_1^2, \quad \int_{-\pi}^{\pi} \frac{d\theta}{\sqrt{2} - \cos \theta} = 2\pi. \quad (12b)$$

This is negative for the values in (12). The mode has zero first-order area and volume changes because its ϕ dependence is $\sin \phi$. Thus the stability assertion cannot be recovered from the cited quadratic functional under the stated conventions. A negative individual coefficient alone would not prove an instability; (12b) instead evaluates the complete quadratic expression for a mode already included in (10). No replacement stability theorem is assumed here.

Area and physical units. For the exact torus, the area is

$$A = 4\pi r_0^2 = 4\pi^2 \sqrt{2} \quad (c = 1). \quad (13)$$

One can evaluate it from the metric without a surface-area formula:

$$\int_{-\pi}^{\pi} \frac{d\theta}{\cosh \eta - \cos \theta} = \frac{2\pi}{\sinh \eta}, \quad \int_{-\pi}^{\pi} \frac{d\theta}{(\cosh \eta - \cos \theta)^2} = \frac{2\pi \cosh \eta}{\sinh^3 \eta}. \quad (13a)$$

The first integral follows on substituting $\tan(\theta/2)$, which turns it into the integral over the real line of $2/[\cosh \eta - 1 + (\cosh \eta + 1) \tan^2(\theta/2)]$ with respect to $\tan(\theta/2)$. Differentiating that result with respect to $\cosh \eta$ gives the second. Multiplying it by $2\pi c^2 \sinh \eta$ gives

$$A = \frac{4\pi^2 c^2 \cosh \eta}{\sinh^2 \eta}, \quad r_0^2 = \pi \sqrt{2} c^2 \quad (\sinh \eta = 1). \quad (13b)$$

Restoring the dimensionless product $c_0 c$ in (12) and using (13b) reproduces the numerical conversion (3); this conversion does not remove the stability gap just identified.

The printed dimensional version of (8) is

$$\Delta p = -2\pi \sqrt{2} \frac{k_c}{r_0^3} c_0 r_0, \quad \lambda = -2 \frac{k_c}{r_0^2} [(2\sqrt{\pi})^{1/2} c_0 r_0 + \frac{1}{4} c_0^2 r_0^2]. \quad (14)$$

The factor $(2\sqrt{\pi})^{1/2}$ in the printed tension formula is inconsistent with (13). Restoring dimensions directly in (8) instead gives

$$\Delta p = -\frac{2k_c c_0}{c^2}, \quad \lambda = -\frac{2k_c c_0}{c} - \frac{1}{2} k_c c_0^2, \quad c = \frac{r_0}{(\pi \sqrt{2})^{1/2}}, \quad (14a)$$

and therefore

$$\lambda = -2 \frac{k_c}{r_0^2} [(\pi\sqrt{2})^{1/2} c_0 r_0 + \frac{1}{4} c_0^2 r_0^2]. \quad (14b)$$

The pressure expression in (14) agrees with (14a). Keeping (14) and (14b) distinct records the discrepancy instead of silently altering the original equation.

3 Gaussian curvature and energy comparison

The extra Gaussian-curvature energy in the paper is

$$F_G = \frac{\bar{k}}{2} \oint c_1 c_2 dA. \quad (15)$$

The factor $1/2$ is part of its convention for $\bar{k}$. Gauss–Bonnet gives

$$\oint K dA = 4\pi(1 - g), \quad F_G = 2\pi\bar{k}(1 - g). \quad (15a)$$

Here g denotes genus, as in this part of the paper; in the earlier coordinate calculations g denoted the metric determinant. At fixed topology F_G is constant, so it changes neither (2) nor the local second variation. The torus has $g = 1$ and the sphere $g = 0$. The zero for the torus can also be checked directly from (10a) and (13a):

$$\oint K dA = 2\pi \int_{-\pi}^{\pi} \left[\frac{1}{(\sqrt{2} - \cos \theta)^2} - \frac{\sqrt{2}}{\sqrt{2} - \cos \theta} \right] d\theta = 0. \quad (15b)$$

The paper next compares a sphere of radius r_0 and a torus of area $4\pi r_0^2$, with common c_0 , Δp , and λ . Substituting $H = -1/r_0$, $K = 1/r_0^2$, and $\nabla^2 H = 0$ into (2), then multiplying by r_0^3 , gives

$$\Delta p r_0^3 + 2\lambda r_0^2 - k_c c_0 r_0 (2 - c_0 r_0) = 0. \quad (16)$$

Indeed, the nonlinear term becomes

$$4k_c(-r_0^{-1} + \frac{1}{2}c_0)(\frac{1}{2}c_0 r_0^{-1})r_0^3 = -k_c c_0 r_0 (2 - c_0 r_0). \quad (16a)$$

Using the scale-consistent relations (14a)–(14b) in (16) cancels the $c_0^2 r_0^2$ terms and leaves

$$-2k_c c_0 r_0 [1 + (\pi\sqrt{2})^{1/2}]^2 = 0. \quad (16b)$$

For $r_0 > 0$ and $k_c > 0$, this means $c_0 = 0$ and consequently $\Delta p = \lambda = 0$. Even the printed factor in (14) gives the same $c_0 = 0$ conclusion, although not the square in (16b). In this argument coexistence means that both geometries satisfy the stationarity equation at the same parameters; equality of their energies is a separate condition.

At $c_0 = 0$, the spherical curvature energy is $\frac{1}{2}k_c(2/r_0)^2(4\pi r_0^2) = 8\pi k_c$. Including the Gaussian term gives the original result

$$F_S = 8\pi k_c + 2\pi\bar{k}. \quad (17)$$

For the torus, scale cancels from $H^2 dA$. At $c = 1$,

$$\oint H^2 dA = \pi \int_{-\pi}^{\pi} \frac{\cos^2 \theta}{(\sqrt{2} - \cos \theta)^2} d\theta = 2\pi^2. \quad (17a)$$

The algebra reducing the integral is

$$\frac{\cos^2 \theta}{(\sqrt{2} - \cos \theta)^2} = \frac{2}{(\sqrt{2} - \cos \theta)^2} - \frac{2\sqrt{2}}{\sqrt{2} - \cos \theta} + 1. \quad (17b)$$

The first two integrals cancel by (13a), leaving 2π . Since $F_G = 0$,

$$F_R = 2k_c \oint H^2 dA = 4\pi^2 k_c. \quad (18)$$

The comparison now consists of a single subtraction:

$$F_R - F_S = 2\pi[2(\pi - 2)k_c - \bar{k}]. \quad (18a)$$

Consequently the ring has the lower energy precisely when

$$\bar{k} > 2(\pi - 2)k_c. \quad (19)$$

Positivity of $\bar{k}$ favors the ring relative to the sphere, but the quantitative threshold is the stronger inequality (19). This compares the two shapes used in the paper, whose volumes differ; it does not by itself establish stability against all deformations or a shape transition at simultaneously fixed area and volume.

The paper's unnumbered discussion also observes that $H = 0$, $c_0 = 0$, and $\Delta p = 0$ satisfy (2) formally. That substitution supplies no closed minimal surface in Euclidean three-space. For a smooth closed Euclidean surface, $\nabla^2 \mathbf{Y} = 2H\mathbf{n}$ and integration by parts would give $\oint |\nabla \mathbf{Y}|^2 dA = 0$ if $H = 0$, which is incompatible with a nondegenerate surface. The discussion of arbitrary genus therefore is not treated here as a reproduced Euclidean existence proof. No further genus calculation is supplied in the original article.

4 The cylinder limit

Returning to (4) with its full scale c gives

$$H = \frac{\sinh^2 \eta + c^2 K}{2c \sinh \eta}, \quad K = \frac{\cosh \eta \cos \theta - 1}{c^2}, \quad \nabla^2 H = -\frac{\cos \theta (\cosh \eta - \cos \theta)^2 \coth \eta}{2c^3}. \quad (20)$$

The powers c^{-1} , c^{-2} , and c^{-3} follow directly from curvature and two surface derivatives. Substitution in (2) restores (6) as

$$\begin{aligned} 0 = \Delta p - 2k_c c^{-2} c_0 (\cosh \eta \cos \theta - 1) - \frac{k_c c_0^2 + 2\lambda}{2c} \left(\sinh \eta + \frac{\cosh \eta \cos \theta - 1}{\sinh \eta} \right) \\ + \frac{k_c}{2c^3} (\cosh \eta - \cos \theta)^2 \left[\frac{\cosh^2 \eta (\sinh^2 \eta - 1)}{\sinh^3 \eta} + \left(\frac{\cosh^3 \eta}{\sinh^3 \eta} - \frac{2 \cosh \eta}{\sinh \eta} \right) \cos \theta \right]. \end{aligned} \quad (21)$$

For finite nonzero c , the cubic coefficient remains $k_c \cosh \eta (1 - \sinh^2 \eta) / (2c^3 \sinh^3 \eta)$. Thus finite rescaling cannot change the selected ratio. A straight cylinder instead requires the singular limit used in the paper:

$$\eta \longrightarrow +\infty, \quad c \longrightarrow -\infty, \quad \frac{c}{\sinh \eta} \longrightarrow -\rho_0, \quad \frac{c}{\cosh \eta} \longrightarrow -\rho_0. \quad (21a)$$

The sign of c implements the orientation convention for comparison with the cylinder in the cited 1989 paper. The generating-circle radius tends in magnitude to ρ_0 , while the distance

of its center from the axis diverges. From (20),

$$\frac{\sinh \eta}{2c} \longrightarrow -\frac{1}{2\rho_0}, \quad \frac{\cosh \eta \cos \theta - 1}{2c \sinh \eta} \longrightarrow 0, \quad \frac{\cosh \eta \cos \theta - 1}{c^2} \longrightarrow 0. \quad (21b)$$

The Laplacian also vanishes: its leading magnitude is proportional to $\cosh^2 \eta / |c|^3$, which tends to zero when $|c| / \cosh \eta \rightarrow \rho_0$. This recovers

$$H = -\frac{1}{2\rho_0}, \quad K = 0, \quad \nabla^2 H = 0. \quad (22)$$

Substituting these into (2), the bending term becomes

$$4k_c \left(-\frac{1}{2\rho_0} + \frac{c_0}{2} \right) \left(\frac{1}{4\rho_0^2} + \frac{c_0}{4\rho_0} \right) = \frac{k_c}{2\rho_0^3} (c_0^2 \rho_0^2 - 1). \quad (22a)$$

The tension term is λ / ρ_0 . Multiplication by ρ_0^3 therefore gives the final equation in the article,

$$\Delta p \rho_0^3 + \lambda \rho_0^2 + \frac{1}{2} k_c (c_0^2 \rho_0^2 - 1) = 0. \quad (23)$$

This is the cylinder's radial equilibrium equation, obtained from the singular limit of (21); condition (7) is not imposed in that limit.

References

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